

Traceability Matrix

ID	Requirements	Related Use Case	Fulfilled By	Tests	Description
1	The application interface contains buttons and displays to simulate a neureset device that works with a PC and a headset	N/A	MainWindow	Run the neureset simulator and observe the UI components included	The neureset system was reproduced using QtCreator's built in user interface and framework. All buttons are clickable by mouse and the application is accompanied by a console log output detailing all actions.
2	The neureset device can be turned on if it has enough battery charge	Use Case #1 Use Case #3	MainWindow Device	Run the simulation. Ensure there is at least 2% battery and press the power button to turn on the device.	The device keeps track of its battery level. Refer to requirement 30 to test turning on the device with low battery.
3	From the main menu, the user can choose to start a new session	Use Case #1	MainWindow Device	Run the simulation and power the device. Notice that the main menu contains three options. Select the 'New session' option.	Using Qt's QPushButton and QListWidget, we can display multiple options on the device screen and use buttons to navigate through the menu.
4	The user can start a new session if the device has enough battery and the headset is connected	Use Case #1	MainWindow Device	Run the simulator and power the device. Ensure a headset is connected by using the headset toggle button.	The device will start a session after verifying that it has enough battery and a headset is connected. Refer to requirement 27 to test replacing the device battery. Refer to 18-20 to test losing headset connection during treatment.

5	User cannot start a new session if there is not enough battery	Use Case #1 Use Case #3	Device	Run the simulation. Set a low battery level using the hardware controls, e.g. 10%. Power the device and try running a new session.	The device checks the battery levels before starting a new session. If it is too low, it won't start a new session.
6	User cannot start a new session if the headset is not connected to the device	Use Case #1	MainWindow Device Headset	Run the simulation and power the device. Use the hardware controls to toggle the headset connection OFF. Try running a new session.	The device checks if a connection from the Headset and Device exists. If the connection is not there, a session cannot be started.
7	A blue light on the device indicates that a headset is connected to it	Use Case #1	MainWindow Device Headset	Run the simulation. Power the device. Press the 'Headset' hardware control button to toggle the connection. Notice the blue light on the top-left corner.	Using a QLabel and stylesheet, we can toggle the lights when the headset connects or disconnects.
8	During a therapy session, the session information is displayed to the user including the approximate time remaining and session progress.	Use Case #1	MainWindow Device Headset	After starting a new therapy session, observe the information displayed on the device screen.	Using various QTWidgets, we are able to display session-related information and update it using signals and slots.
9	During a therapy session, record the starting frequencies at	Use Case #1	Device Headset SessionLog	After starting a new therapy session, observe the session information being displayed on the	For each site, the Headset generates a random waveform from 4 out of 5 frequency bands: alpha [8-12Hz], beta [12-30Hz], delta [1-4Hz], theta [4-8Hz],

	different wavelengths for every node on the headset.			console during the session. Once completed, the information is available in a session log.	and gamma [25-140Hz]. The frequency and amplitude of the waveform before treatment is stored in the session log.
10	During a therapy session, display a sample waveform graph from one of the EEG sites.	Use Case #1	MainWindow Device Headset	After starting a new therapy session, observe the graph displayed in the device screen.	The device will choose a random EEG site's waveform and graph the sum of its components as a single wave in the device screen.
11	During a therapy session, before treatment, calculate a dominant frequency using the frequencies and amplitudes of each EEG site.	Use Case #1	Device Headset	After starting a new therapy session, observe the dominant frequency displayed in the console. Once completed, the information is also available in a session log.	The device uses the waveforms generated by the headset to perform the dominant frequency calculations.
12	During a therapy session, treatment starts by applying a frequency equal to the dominant frequency plus an offset to each node, every 1/16th of a second for the duration of one second	Use Case #1	Device Headset	Run a new session on the neureset simulator and observe the console log output. Check the session logs after uploading them on the PC and notice that the different frequencies applied to each site for every round.	After calculating a dominant frequency, the treatment frequency is calculated and applied with the Headset. A QTimer is used to measure the treatment intervals.
13	During a therapy	Use Case #1	Device	Run a new session on	The treatment frequencies are stored in a

	session, treatment frequencies are stored in the session log.		Headset SessionLog	the neureset simulator and observe the console log output. Check the session logs after uploading them on the PC and notice that the different frequencies applied to each site for every round.	QVector in the session log object.
14	The treatment frequency offset ranges from 5 to 20 Hz and is increased by 5Hz for each treatment round (for a total of 4 rounds)	Use Case #1	Device Headset	Run a new session on the neureset simulator and observe the frequency applied at each node in the console log output. Check the session logs after uploading them on the PC and notice that the increase of 5 Hz in frequencies applied at every round.	We increment the offset by 5 after each round, and we send the frequency to the headset to apply the correct frequencies
15	During a treatment round, a green light flashes indicating that treatment is being applied	Use Case #1	MainWindow Device Headset	Run a new session on the simulator. Notice that the green light on the device turns green during each of the 4 rounds of treatment.	Using a QLabel and stylesheet, we can intermittently toggle the lights while the treatment rounds are being applied.
16	During a therapy session, record the baselines after treatment rounds for every node on the headset.	Use Case #1	Device Headset SessionLog	After starting a new therapy session, observe the session information being displayed on the console during the session. Once	Similar to requirement 9. The frequency and amplitude of the waveforms after treatment is stored in the session log.

				completed, the information is available in a session log.	
17	After the therapy session, a session log is saved.	Use Case #1	Device SessionLog	Run a new session and once it is finished, check the session information printed on the console. The log can be viewed on the session log menu or PC once uploaded.	The session log that was saved is stored in a vector of session logs which can save multiple session logs.
18	During a therapy session, if the headset disconnects from the device, pause the session.	Use Case #1	MainWindow Device Headset	After starting a new session, use the hardware control buttons to toggle the Headset connection. Observe that the session is paused.	This is implemented using signals and keeping track of the state upon which the session was paused.
19	During a therapy session, if the headset disconnects from the device, a red light flashes and the device starts beeping	Use Case #1	MainWindow Device Headset	After starting a new session, use the hardware control buttons to toggle the Headset connection. Observe that the session is paused. Observe the red light in the top-left corner. Observe the console outputs for a beeping message.	This is implemented using signals and keeping track of the state upon which the session was paused. Using a QLabel and stylesheet, we can intermittently toggle the red light while the headset is disconnected. The sound is simulated with console outputs.
20	During a therapy session, if the headset disconnects from the device and	Use Case #1	MainWindow Device Headset	After starting a new session, use the hardware control buttons to toggle the Headset connection. Observe that	This is implemented using signals and keeping track of the state upon which the session was paused.

	then reconnected, resume the session			the session is paused. Toggle the button again. Observe that the session resumes.	
21	During a therapy session, the user can press the pause/resume button on the device to pause and resume a session.	Use Case #1	MainWindow Device	Run a new session on the neureset simulator and click the pause/resume button. Notice the progress messages displayed as you resume or pause the session.	Using a QPushButton, we can call the function that pauses the session by pressing the pause button. If the button is pressed again, it will resume the session.
22	If a session is paused for longer than 5 minutes, the device will turn off and erase the current session log.	Use Case #1	MainWindow Device	Run a new session on the neureset simulator and pause the session/toggle the headset connection. Wait until the timer times out and notice that the device turns off. Turn the device back on and notice that the session does not appear in the session logs.	When a session is paused, the device starts a timer that will stop the current session after a set interval. The testing interval is 15 seconds.
23	The user can update the current date and time on the device	Use Case #2	MainWindow Device	Run the simulation. Power the device. Select the 'Time & Date' option from the home menu. Change the time and/or date and press the OK button. The change is reflected in subsequent session logs. This can be	The device has a QDateTime that can be updated to the date specified by the user.

				verified by running a new session and checking the starting time.	
24	Device battery is shown at all times	Use Case #3	MainWindow Device	Run the simulation. Observe the battery icon on the top-right corner of the device.	The device keeps track of the battery levels with a variable.
25	When the device detects low battery, a low battery icon is displayed in the corner of the screen	Use Case #3	Device MainWindow	Run the simulation. Use the hardware controls to set a low battery, observe how the battery icon changes colors.	Using the member variable in the Device class, we can track the battery percentage. Each time that the battery is updated, it checks if it needs to update the icons based on fixed thresholds. Below 10% is very low, shown as red. Below 20% is low, shown as yellow. Normal levels are green.
26	When the device detects low battery, a notification is displayed. The notification can be dismissed.	Use Case #3	Device MainWindow	Run the simulation. Use the hardware controls to set a low battery, at around 20% the notification should show up.	The device will signal the main window once the battery reaches a threshold so that it displays the notification.
27	The user can replace a low battery by turning off the device and inserting new batteries.	Use Case #3	MainWindow Device	Run the simulation. Use the hardware control to set the battery to anything below 20%. Turn on the device. Observe the low battery warnings (or device shutdown if battery is 0%). Turn off the device. Use controls to increase	The device uses signals to notify the MainWindow about the low battery levels. The MainWindow hardware controls simulate the user sliding off the battery cover and replacing the physical batteries.

				battery level. Turn on the device. The warnings should no longer be there.	
28	If the device detects low battery during a session, it tries to finish the session but is unable to start a new session until the battery is replaced.	Use Case #1/ Use Case #3	Device	Run the simulation. Set the battery threshold of 45% and start a new session. Notice that the session finishes and the logs are saved properly but a new session with low battery cannot be started.	Using the member variable in the Device class, we can track the battery percentage. Each time that the battery is updated, it checks if it is able to continue the session. If it is, it will finish the current session. One session uses approximately 30% of the battery.
29	During a session, if the device battery levels are too low, the session is aborted	Use Case #3	Device MainWindow	Run the simulation. Set a battery threshold of 30% and start a session. The session should pause once the battery is too low.	The device uses signals to check battery levels at every update of the battery.
30	If the batteries are completely depleted, the device shuts down automatically	Use Case #3	MainWindow Device	Run the simulation. Turn on the device. Use the hardware controls to set the battery level to 0. Wait 2 seconds for the device to shut down.	The device uses signals to check battery levels at every update of the battery. The device sends a power down signal once the battery levels are too low.
31	From the main menu, the user can choose to view session logs in the device	Use Case #4	MainWindow Device	Run the simulation and power the device. Notice that the main menu contains three options. Select the 'Session log' option.	Using Qt's QPushButton and QListWidget, we can display multiple options on the device screen and use buttons to navigate through the menu.

32	The device can be connected to a PC	Use Case #4	MainWindow Device PC	Run the simulation and power the device. Use the hardware controls to toggle the PC connection ON.	The device keeps track of the PC connection with a variable.
33	The user can upload session logs from the device to the PC	Use Case #4	MainWindow Device PC	Run a therapy session. Navigate to the session logs and press the 'upload logs' button. The PC screen should display a list of uploaded logs.	The SessionLog objects stored in the device are pushed to a vector of SessionLogs in the PC.
34	The user can view detailed information about a session using the PC. Information includes before and after baselines, date & time, per site information and more.	Use Case #4	MainWindow Device PC	After uploading some session logs to the PC, double click one of the logs in the PC screen to view detailed information.	The PC screen has a second window to display detailed session information. The information is read from the session log chosen.